

Compare Results

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USB_PD_R3_2 V1.0 2023-10_Part24.pdf

6 pages (737 KB)

15/04/2024 10:53:00

versus

New File:

USB_PD_R3_2 V1.1 2024-10_Appendix D BMC.pdf

6 pages (1.39 MB)

28/10/2024 11:27:53

Total Changes

116

Text only comparison

Content

- 114 Replacements
- 0 Insertions
- 2 Deletions

Styling and Annotations

- 0 Styling
- 0 Annotations

Go to First Change (page 1)

D BMC Receiver Design Examples

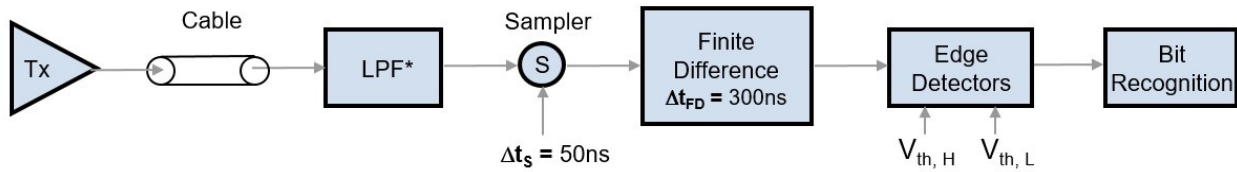
The *BMC* signal is DC-coupled so that the voltage level is affected by the ground *IR Drop*. The DC offset of the *BMC* signal at Power Source and Power Sink are in the opposite directions. When the *V_{BUS}* current is increased from 0A, the *BMC* signal waveform shifts downward at Power Sink and shifts upward at Power Source. This section introduces two sample *BMC* receiver circuit implementations, which are immune from DC offset and high current load step. They can be used in Power Source, Power Sink and inside cables.

D.1 Finite Difference Scheme

D.1.1 Sample Circuitry

The sample Finite Difference *BMC* receiver shown in [Figure D.1, "Circuit Block of BMC Finite Difference Receiver"](#) consists of the Rx bandwidth limiting filter with the time constant *t_{RxFilter}*, a sampler with the sampling step Δt_s , 50ns, a Finite Difference Calculator which calculates the voltage difference between the time interval of Δt_{FD} , 300ns, an edge detector controlled by two voltage thresholds, $V_{th,H}$ and $V_{th,L}$ and a logic block for bit recognition.

Figure D.1 Circuit Block of BMC Finite Difference Receiver



D.1.2 Theory

This section describes the fundamental theory of Finite Difference Scheme to recover the received *BMC* signal with the input and output signal waveforms of the circuit blocks shown in [Figure D.1, "Circuit Block of BMC Finite Difference Receiver"](#). To illustrate the robustness of the implementation, the *V_{BUS}* current load step rate is intentionally increased to 2A/μs at the Sink load. In [Figure D.2, "BMC AC and DC noise from V_{BUS} at Power Sink"](#) (a), the red curve represents the *V_{BUS}* current measured at the Power Sink when the current is increased at 9 μs from 0A to 5A and the blue dash curve represents the *V_{BUS}* current measured at the USB Type-C® connector of the power Sink. In this example, the peak current overshoot with larger load step rate is increased to 518 mA which exceeds *iOvershoot*. [Figure D.2, "BMC AC and DC noise from V_{BUS} at Power Sink"](#) (b) shows the total *BMC* noise at Power Sink, coupled from *V_{BUS}* and D+/D- through the worst [USB Type-C 2.4] compliant cable, after the Rx bandwidth limiting filter with the time constant *t_{RxFilter}* is applied. The noise can be decomposed into 3 components. The first is the DC offset, $I_{VBUS}(t) \cdot R_{GND}$, while I_{VBUS} is the *V_{BUS}* current and R_{GND} is the ground DC resistance of the cable. The offset is negative in Power Sink and positive at Power Source. The second noise component is the inductive *V_{BUS}* noise, $M \cdot dI_{VBUS}(t)/dt$, while M is the mutual inductance between the *V_{BUS}* and CC wires in the cable and $dI_{VBUS}(t)/dt$ is the load step rate. The third component is [USB 2.0] Full Speed SE0 coupling noise which would normally occur randomly but was assumed to occur periodically in the simulation to account for the crosstalk in any phase between the *BMC* and [USB 2.0] signals. In [Figure D.3, "Sample BMC Signals \(a\) without USB 2.0 SE0 Noise \(b\) with USB 2.0 SE0 Noise"](#), the blue dash curve represents the *BMC* signal when there is no *V_{BUS}* current, and the red solid curve represents the *BMC* signal affected by the *V_{BUS}* coupling noise shown in [Figure D.2, "BMC AC and DC noise from V_{BUS} at Power Sink"](#) (b). The green solid curve is the sample [USB 2.0] noise, after the Rx bandwidth limiting filter with the time constant *t_{RxFilter}* is applied.

Figure D.2 BMC AC and DC noise from VBUS at Power Sink

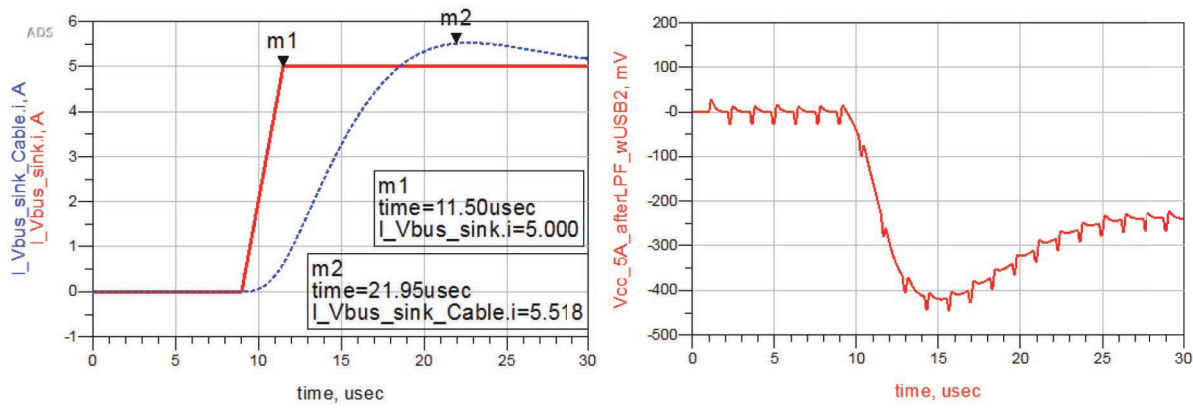
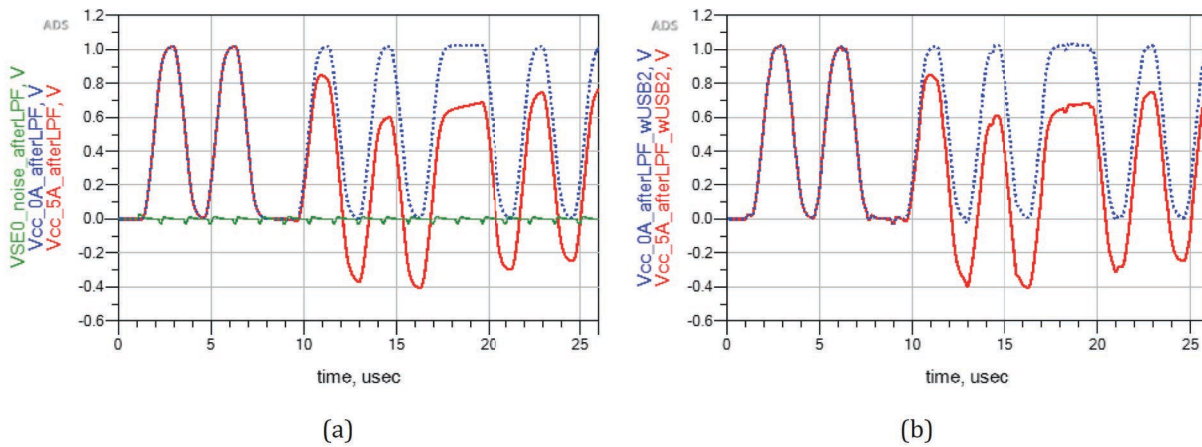
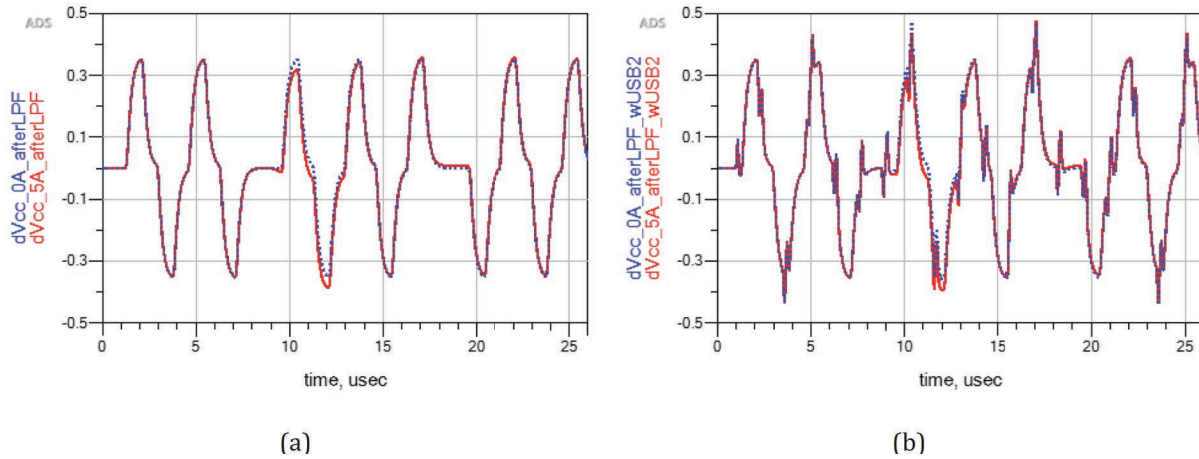


Figure D.3 Sample BMC Signals (a) without USB 2.0 SE0 Noise (b) with USB 2.0 SE0 Noise



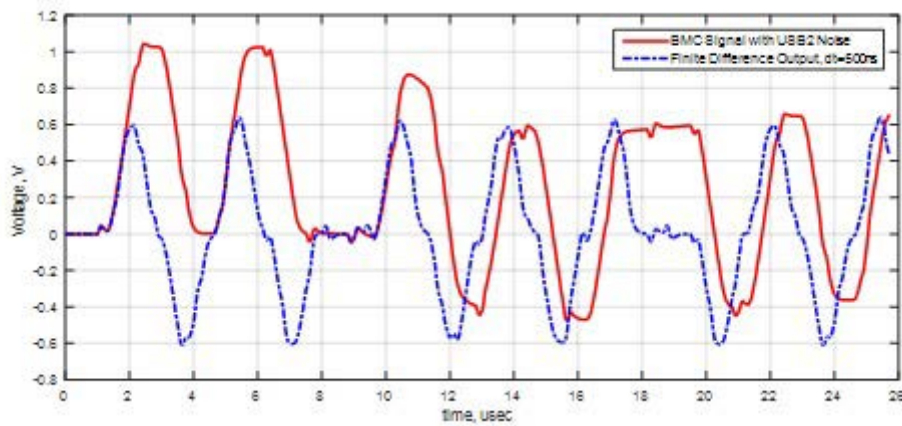
The BMC signals shown in [Figure D.3, "Sample BMC Signals \(a\) without USB 2.0 SE0 Noise \(b\) with USB 2.0 SE0 Noise"](#) are sampled every 50ns and the scaled derivative waveforms, $V_{cc}(t) - V_{cc}(t - 50ns)$, without and with [\[USB 2.0\]](#) noise are shown in [Figure D.4, "Scaled BMC Signal Derivative with 50ns Sampling Rate \(a\) without USB 2.0 Noise \(b\) with USB 2.0 Noise"](#) (a) and (b), respectively. In [Figure D.4, "Scaled BMC Signal Derivative with 50ns Sampling Rate \(a\) without USB 2.0 Noise \(b\) with USB 2.0 Noise"](#) (a), if there is no [\[USB 2.0\]](#) noise, the derivative waveform just changes slightly before and after the V_{BUS} current transition. That means, the slope of the BMC waveform is not sensitive to the DC offset and is very useful to be used to design a robust receiver against a large DC offset. However, the derivative waveforms with [\[USB 2.0\]](#) noise have large perturbation as shown in [Figure D.4, "Scaled BMC Signal Derivative with 50ns Sampling Rate \(a\) without USB 2.0 Noise \(b\) with USB 2.0 Noise"](#) (b).

Figure D.4 Scaled BMC Signal Derivative with 50ns Sampling Rate (a) without USB 2.0 Noise (b) with USB 2.0 Noise



To remove the high frequency content of the **[USB 2.0]** noise, Finite Difference technique with the proper time interval is applied to the **BMC** waveform with **[USB 2.0]** noise in [Figure D.3, "Sample BMC Signals \(a\) without USB 2.0 SE0 Noise \(b\) with USB 2.0 SE0 Noise"](#). Using Backward Finite Difference Calculator, $\Delta V_{cc} = V_{cc}(t) - V_{cc}(t - \Delta t)$, [Figure D.5, "BMC Signal and Finite Difference Output with Various Time Steps"](#) shows the Finite Difference Output while $\Delta t = 500\text{ns}$. The larger the time interval Δt is, the larger the peak-to-peak magnitude of the Finite Difference Output will be. However, the time interval is bounded by the rise time of the **BMC** signal so that 300ns to 500ns is a good range of the time interval.

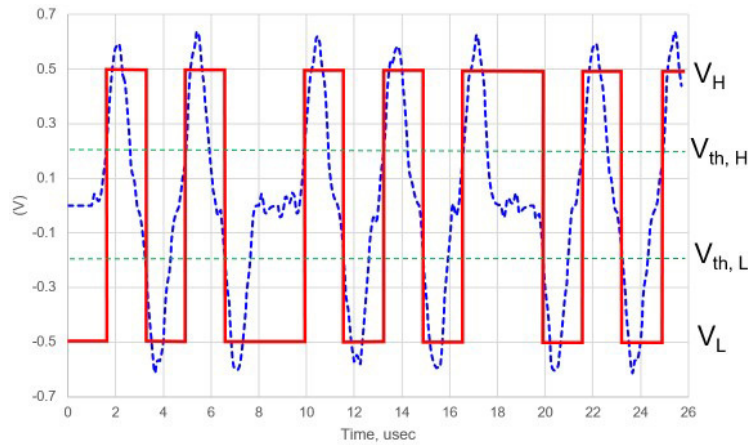
Figure D.5 BMC Signal and Finite Difference Output with Various Time Steps



D.1.3 Data Recovery

The edge detection is followed by the Finite Difference Calculation. At the input of the edge detector, if the voltage is larger than $V_{th,H}$ at the rising edge, the output will become high voltage level, V_H , if the voltage is smaller than $V_{th,L}$ at the falling edge, the output will become low voltage level, V_L . In this example, $V_{th,H}$ and $V_{th,L}$ are 0.2V and -0.2V, respectively. The solid curve in [Figure D.6, "Output of Finite Difference in dash line and Edge Detector in solid line"](#) represents the output of the edge detector, where V_H is 0.5V and V_L is -0.5V.

Figure D.6 Output of Finite Difference in dash line and Edge Detector in solid line



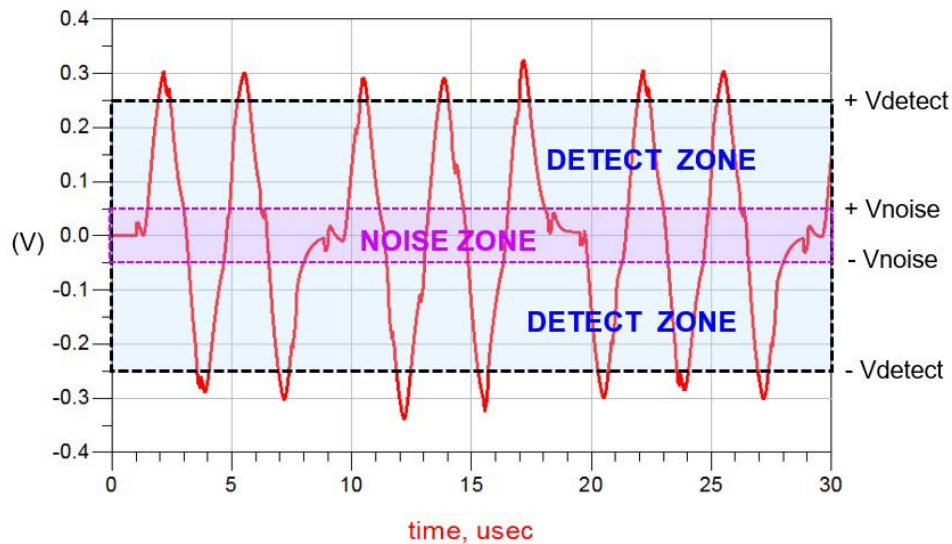
The duty cycle of the output signal from the edge detector varies depending on the thresholds, $V_{th,H}$ and $V_{th,L}$, as well as jitter and noise from silicon and channel. The techniques such as integrating receiver can be used to recover the *BMC* signal.

D.1.4 Noise Zone and Detection Zone

Figure D.7, "Noise Zone and Detect Zone of BMC Receiver" shows the output of Finite Difference when the time interval of Finite Difference is set to 300ns. The noise Zone is defined in between $+V_{noise}$ and $-V_{noise}$, in which the noise glitches occur. The detect zone is defined in between $+V_{detect}$ and $-V_{detect}$, excluding the noise zone. The thresholds of the edge detectors, $V_{th,H}$ and $V_{th,L}$, must be properly set within the detect zone so that the data can be recovered successfully.

In this example, V_{detect} is 250mV and V_{noise} is 50mV. It is highly recommended that the product implemented with the similar techniques indicates the performance with the range of V_{noise} and V_{detect} in the electrical specification.

Figure D.7 Noise Zone and Detect Zone of BMC Receiver

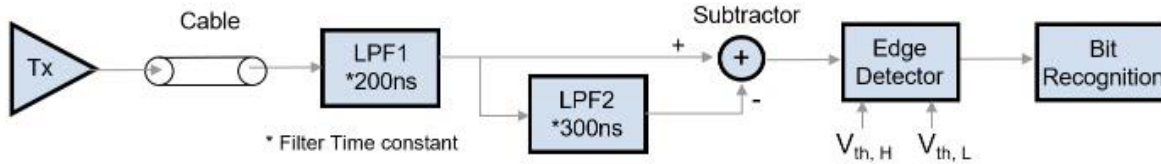


D.2 Subtraction Scheme

D.2.1 Sample Circuitry

The sample Subtraction BMC receiver shown in [Figure D.8, "Circuit Block of BMC Subtraction Receiver"](#) consists of the two Low Pass Filters (LPF1 and LPF2), a Subtractor, an Edge Detector and a logic block for bit recognition. The time constant of the first and second LPF are 200ns and 300ns, respectively. The Subtractor subtracts the LPF1 output from the LPF2 output. The Edge Detector controlled by two voltage thresholds, $V_{th,H}$ and $V_{th,L}$ to recover the data.

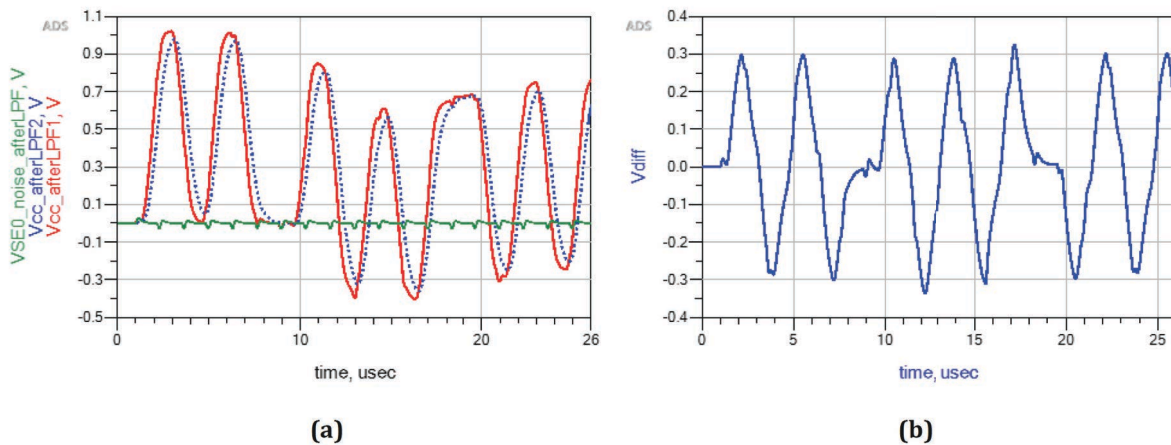
Figure D.8 Circuit Block of BMC Subtraction Receiver



D.2.2 Output of Each Circuit Block

[Figure D.9, "\(a\) Output of LPF1 and LPF2 \(b\) Subtraction of LPF1 and LPF2 Output"](#) (a) shows the output of LPF1 as the red solid line and LPF2 as the blue dash line as well as the [\[USB 2.0\]](#) noise in green solid line. [Figure D.9, "\(a\) Output of LPF1 and LPF2 \(b\) Subtraction of LPF1 and LPF2 Output"](#) (b) shows the voltage difference between the two output filters, $V_{diff} = V_{cc_afterLPF1} - V_{cc_afterLPF2}$. The V_{diff} waveform looks very similar to the Finite Difference output waveform shown in [Figure D.6, "Output of Finite Difference in dash line and Edge Detector in solid line"](#) so that the data recovery method through the edge detector is the same as described in [Section D.1.3, "Data Recovery"](#).

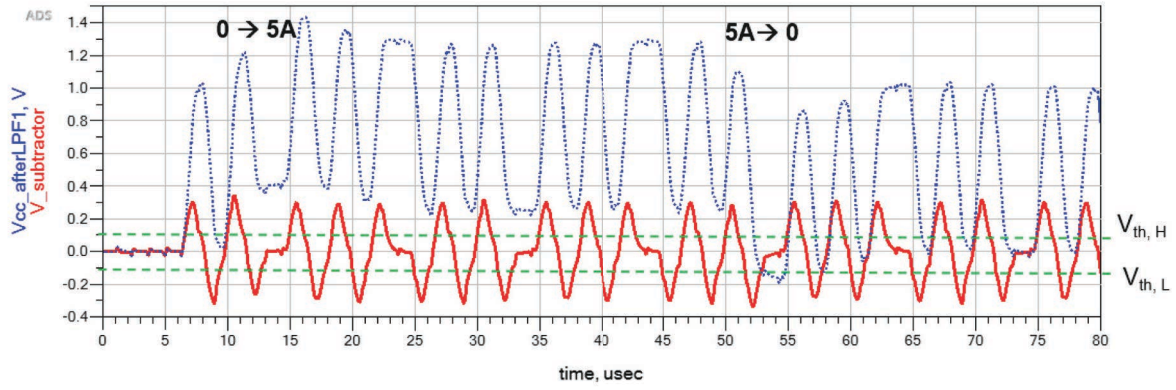
Figure D.9 (a) Output of LPF1 and LPF2 (b) Subtraction of LPF1 and LPF2 Output



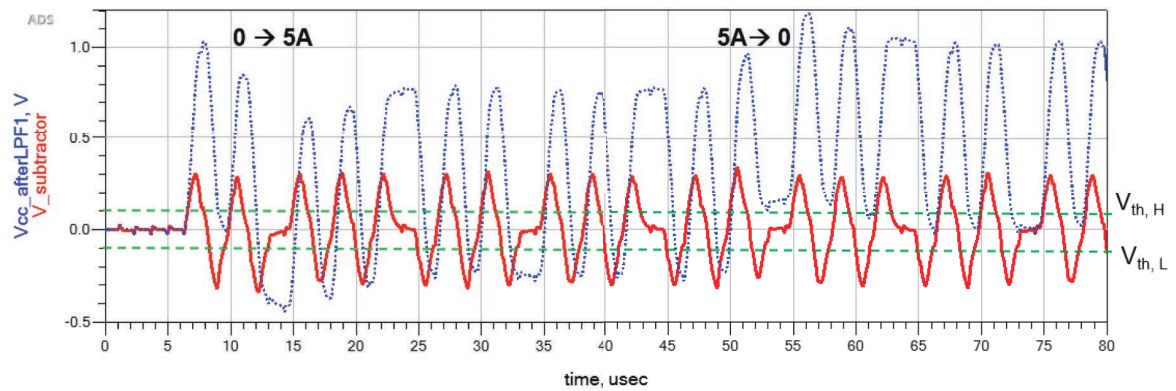
D.2.3 Subtractor Output at Power Source and Power Sink

The following figures show the example when the V_{BUS} current increases from 0A to 5A and then decreases to 0A with high load step rate. The output of the LPF1 and the Subtractor at Power Source and Power Sink are shown in [Figure D.10, "Output of the BMC LPF1 in blue dash curve and the Subtractor in red solid curve \(a\) at Power Source \(b\) at Power Sink"](#) (a) and (b), respectively. Although the BMC signals at Power Source and Power Sink shift toward the opposite direction, the Subtractor outputs at Power Source and Power Sink are almost identical regardless of the opposite direction of the DC offset.

Figure D.10 Output of the BMC LPF1 in blue dash curve and the Subtractor in red solid curve (a) at Power Source (b) at Power Sink



(a)



(b)

D.2.4 Noise Zone and Detection Zone

The zone definition is the same as defined in [Section D.1.4, "Noise Zone and Detection Zone"](#). The sizes of the noise zone and detection zone of the Subtraction Scheme are dependent on the filter time constant. When the time constant of the first and second LPF are 200ns and 300ns, respectively, V_{detect} is 250mV and V_{noise} is 50mV. It is highly recommended that the product implemented with the similar techniques indicates the performance with the range of V_{noise} and V_{detect} in the electrical specification.